Matthew Laws

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EDUCATION

Northeastern University, Boston, MA

Doctor of Philosophy, Computer Science, Expected April 2030

Williams College, Williamstown, MA

GPA: 3.96 (Major) 3.91 (All), Bachelor of Arts, Highest Honors in Computer Science, Cum Laude, June 2025

Selected Coursework: Operating Systems, Distributed Systems, Parallel Processing, Deep Learning, Natural Language Processing, Machine Learning, Causal Inference, Probability, Linear Algebra, Algorithms & Analysis, Data Structures.

Honors: Sigma Xi, Goldberg Prize (Best Senior Thesis in CS), CRA Undergraduate Researcher Award: Honorable Mention, Williams College x Mysten Labs Hackathon: 1st Place, Class of 1960 Scholar of Computer Science, Ward Prize Nomination

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Go, JavaScript (Proficient) | R, Java, Shell, SQL, F# (working knowledge)

Tools: React, AWS S3 / & R53, Node, Git, Ab Initio, CSS (Development) | Snowflake, Postgres (Database) | Torch, Tensorflow (Keras), Scikit-learn, OpenCV, Hydra (AI/ML) | OpenGL, GLSL, x86, LaTeX, .NET (Other)

PUBLICATIONS AND PRESENTATIONS

SPIDER: Improved Succinct Rank and Select Performance (Publication)

M. D. Laws, J. Bliven, K. Conklin, E. Laalai, S. McCauley, and Z. S. Sturdevant | Symposium on Experimental Algorithms (SEA), 2024. [arXiv] [LIPIcs]

Causal Inference with Protein Sequences: Biologically Plausible Dimension Reduction Using AlphaFold (Presentation)

M. D. Laws, R. Bhattacharya | American Causal Inference Conference (ACIC), 2025

SELECTED PROJECTS

Causal Proteins: Developed a system for determining the causal effect that proteins have on disease. Achieved 91% identification precision for pathogenic diseases. Uses parallel GPU processing for efficiency.

Peer-to-Peer Federated Learning: Developed a system using Go and Python for a P2P federated learning system.

ASL Interpreter: Developed a computer vision model that identifies ASL letters. Attained 97% accuracy for letters and 95% accuracy for words. Ships with a live video recognition feature for real time translation.

SOFTWARE EXPERIENCE

Williams College, Williamstown, MA, *Post-baccalaureate Research Assistant* June 2025 - Present • Generate a publication-ready manuscript • Build a software package for causal inference with proteins • Report weekly to Professor Bhattacharya and collaborators at Columbia University and Johns Hopkins University.

Sanofi, Cambridge, MA, *Data Scientist Intern* May 2024 - August 2024 • Worked in the ML / AI team as a Data Scientist to leverage computer vision techniques in the clinical setting • Summarized findings in a paper currently under company review.

Williams College, Williamstown, MA, *Teaching Assistant* Sept 2022 - June 2025 • Hosted multiple TA sessions a week to help students with lab assignments • Past courses: Robotics, Deep Learning, Computer Organization (3 Semesters), Data Structures.

Williams College, Williamstown, MA, *Computer Science Research Fellowship* June 2023 - May 2024 • Worked with Professor Sam McCauley to develop a faster Succinct Rank and Select Data Structure • Developed theoretical algorithms, implemented, and optimized them in C • Presented my research in a talk at the Symposium on Experimental Algorithms 2024.

Integral Ad Sciences, New York, NY, *Data Engineering Intern* June 2022 - August 2022 • Worked in a team of 4 using the Agile methodology to develop a regression testing software IAS's twitter brand safety pipeline.

LEADERSHIP EXPERIENCE

Williams Students Online (Student Website): *President,* June 2024 - June 2025

Williams College Rugby Football Club (WRFC): *Captain,* Sept 2021 - May 2025

Williams Record, *Website Editor* Jan 2024 - May 2025

Williams College Junior Advisor Program: *Junior Advisor,* May 2023 - May 2024